Amazon Book Analytics

Pricing, Trends & Historical Context

Q1 How did book and category popularity evolve across decades?

Q2 Which categories command premium prices, and how stable are they?

Q3 How does author gender distribution vary across categories and correlate with pricing?

Riddhi Kedia · Data Science Foundations — Midterm Assignment · January 2026

DATA FOUNDATION

Data Sources & Pipeline Architecture

212k+

Books

Amazon Books Dataset

Kaggle

Titles, categories, prices, publication years, authors

480k+

Reviews

Amazon Reviews Dataset

Kaggle

Ratings and review counts — used to build popularity scores

Top 30

Authors

Genderize.io API

External Integration

First-name gender inference for author gender analysis

PIPELINE

Data Acquisition → Cleaning & Standardisation → **DuckDB Architecture** → SQL-First Analysis → Visualisation & Insights

DuckDB as the analytical core — columnar storage optimised for fast aggregations across 700k+ records using structured SQL joins across 3 tables.

Analytical Approach

01 Data Preparation

- Cleaned prices (numeric enforcement) and review scores (removed outliers outside 0–5)
- Extracted year/decade, author first name, and primary category from raw fields
- Outlier detection and data validation before any analysis

02 External API Integration

- Called Genderize.io API on top 30 authors by publication volume
- Inferred author gender from first name; joined results back to books dataset
- Documented confidence thresholds and API rate limit handling

03 DuckDB Architecture

- Built 3-table schema: Books, Reviews, Author Genders
- Structured joins to enable cross-dataset queries at scale
- All downstream analysis executed via SQL — no pandas aggregations post-load

04 Analysis & Visualisation

- Popularity scoring: weighted review volume × average rating, segmented by decade
- Price stability: Coefficient of Variation (CV) weighted by sample size per category/decade
- Gender pricing: statistical significance tested via p-values to separate signal from noise

Most Popular Book Per Decade (1900s–2020s)

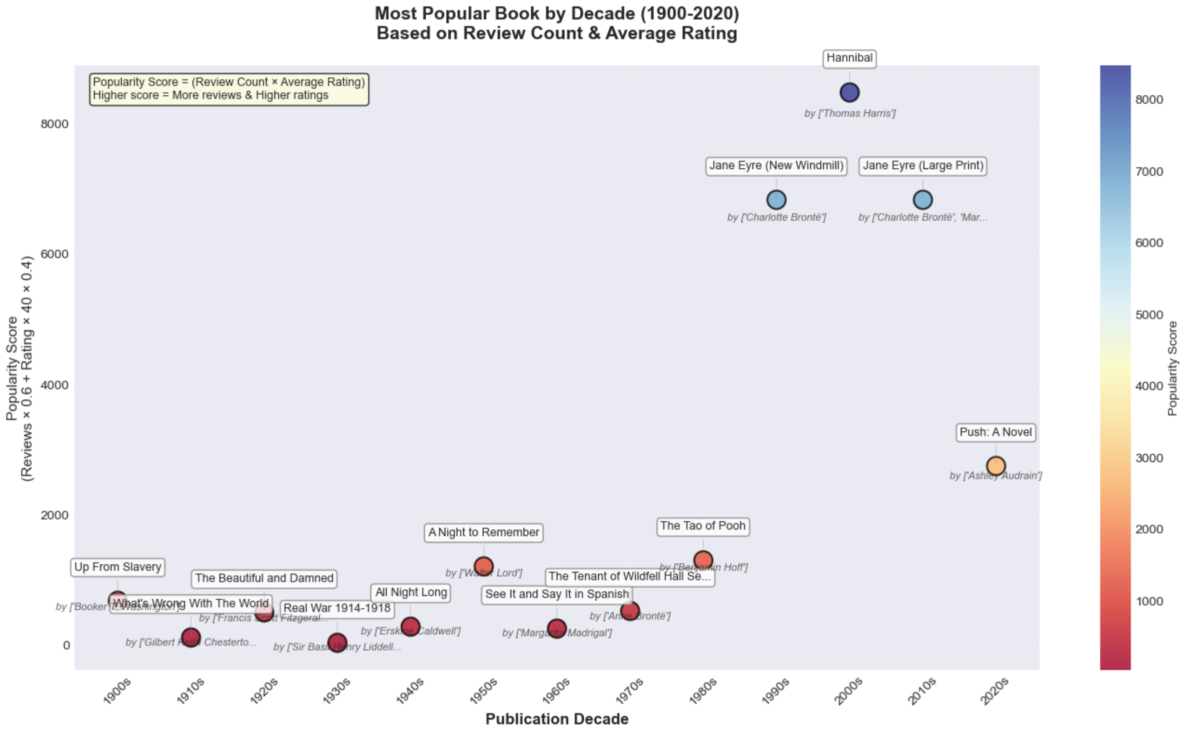
Decade	Title	Author(s)	Score	Reviews	Rating
1900s	*Up From Slavery*	Booker T. Washington	677.0	151	4.48
1910s	*What's Wrong With the World*	Gilbert Keith Chesterton	112.0	25	4.48
1920s	*The Jungle Book*	Rudyard Kipling	1661.0	381	4.36
1930s	*The Crusades: The World's Debate*	Hilaire Belloc	33.0	7	4.71
1940s	*All Night Long*	Erskine Caldwell	279.0	66	4.23
1950s	*A Night to Remember*	Walter Lord	1205.0	264	4.56
1960s	*Morte Darthur (York Medieval Texts)*	Thomas Malory	320.0	73	4.38
1970s	*The Tenant of Wildfell Hall Set*	Anne Brontë	521.0	116	4.49
1980s	*The Tao of Pooh*	Benjamin Hoff	1298.0	304	4.27
1990s	*Jane Eyre (New Windmill)*	Charlotte Brontë	6839.0	1523	4.49
2000s	*Hannibal*	Thomas Harris	8489.0	2793	3.04
2010s	*Jane Eyre (Large Print)*	Charlotte Brontë; Marc Cac…	6839.0	1523	4.49
2020s	*Push: A Novel*	Ashley Audrain	2748.0	661	4.16

KEY INSIGHTS

- Post-1990 books have 10× more reviews — digital reviewing transformed what 'popular' means in the dataset.
- Hannibal tops the 2000s despite only 3.04 stars — review volume outweighs quality in the composite score.
- Jane Eyre appears twice (1990s and 2010s) — rare evidence of a classic sustaining genuine modern engagement.
- Popularity ≠ Quality: the scoring methodology intentionally weights market reach over rating.

QUESTION 1 · POPULARITY

Category Popularity Trends by Decade



PEAK BY DECADE

1900s Fairy Tales

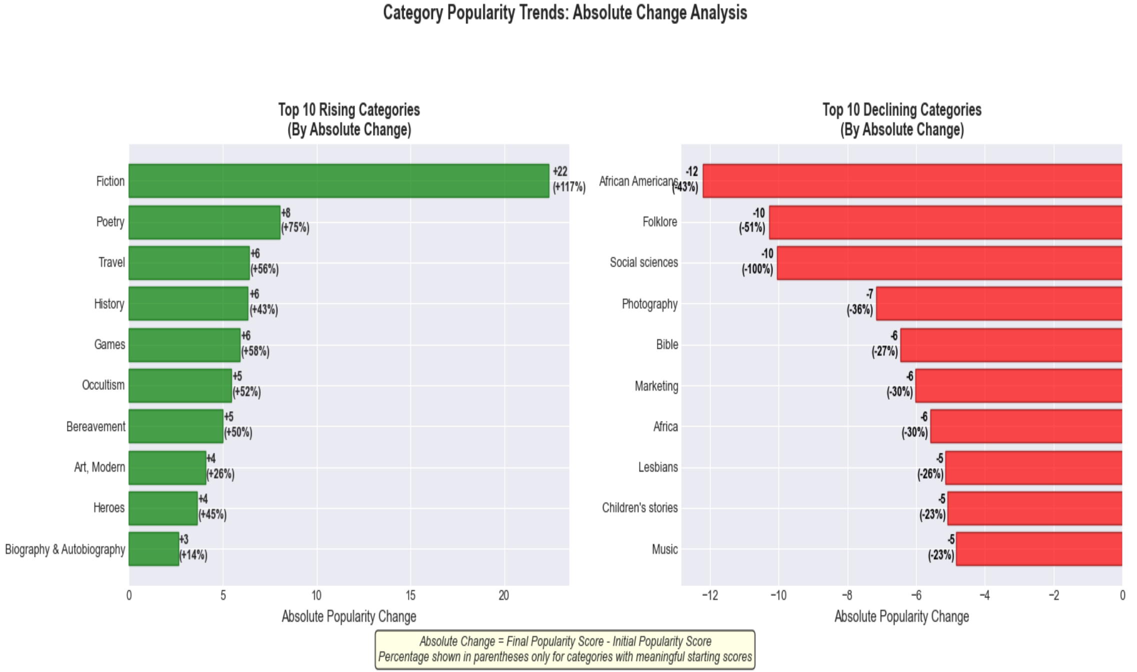
1920s Animals

1930s–60s Fiction dominates

1970s Juvenile Fiction

1980s–2020s Fiction

Category Trends: Gainers, Decliners & Stable Categories



KEY INSIGHTS

- Fiction: biggest absolute gainer (+22.36), from 19.15 → 41.50 across the century.
- African Americans category declined most sharply (−12.21), prominent early 20th century, faded since.
- Science: most stable over 80 years — score barely moved (17.23 → 17.26, avg rating 4.23★).
- 20 categories grew, 30 stable, only 5 declined sharply — the market is broadly stable.

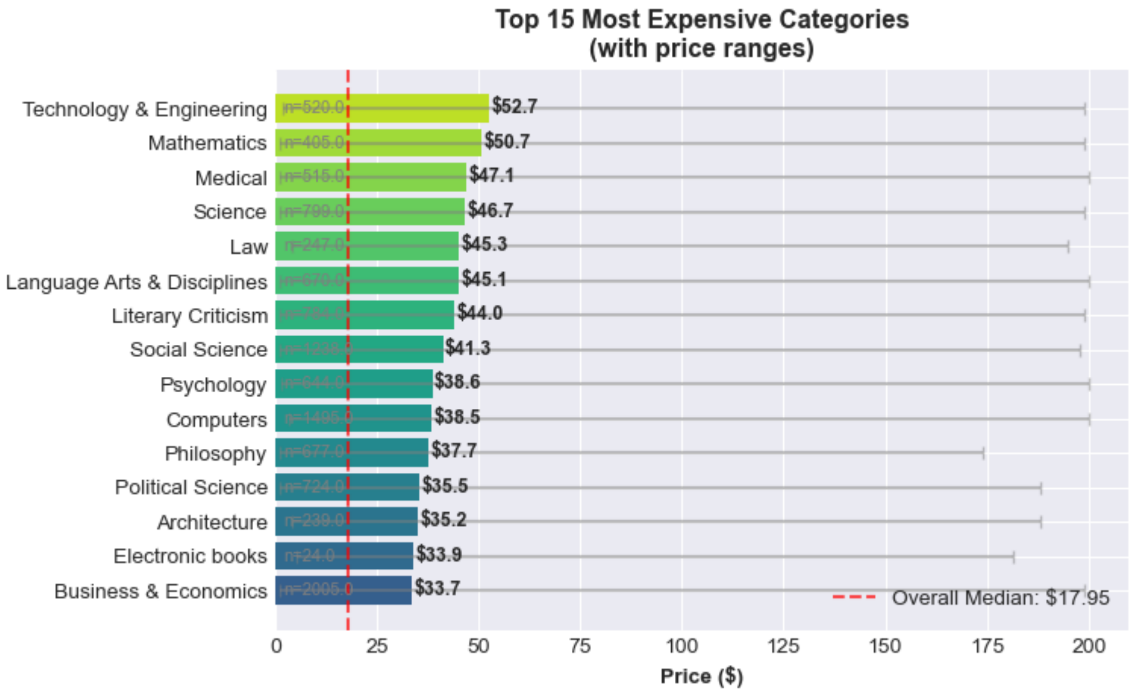
20
Growing categories

30
Stable categories

5
Declining categories

Stability is the norm — growth or sharp decline is the exception.

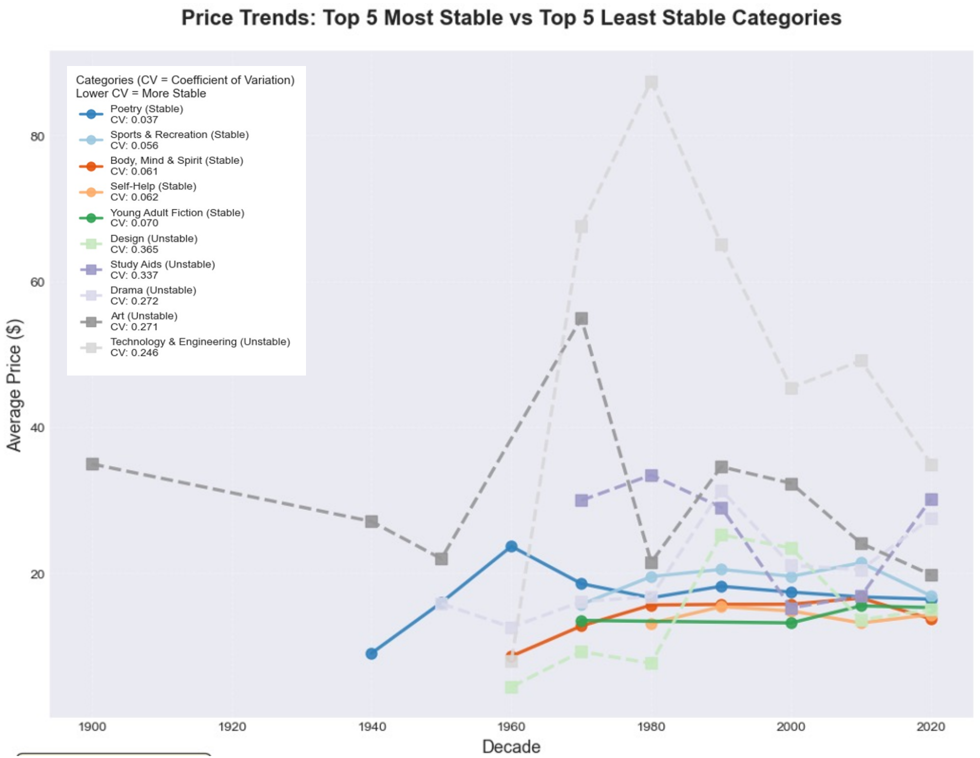
Which Categories Command Premium Prices?



KEY INSIGHTS

- Technical/academic categories lead: Technology & Engineering ($52.54), Mathematics ($50.88), Medical ($47.07) — all 3× the price of leisure content.
- Children's content is consistently affordable: Juvenile Fiction ($12.09), Young Adult Fiction ($14.84).
- Leisure categories cluster $13–$17 — Crafts, Travel, Humor, Comics all show established price conventions.
- Overall median $17.95 — premium technical categories are clear outliers, not the norm.

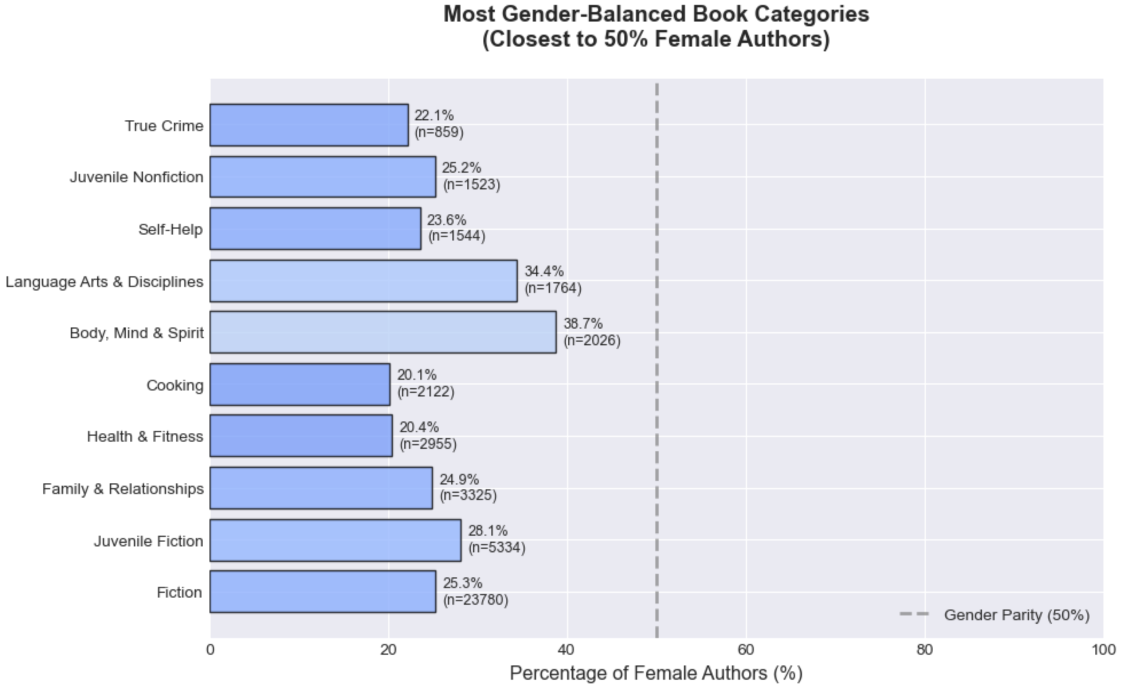
Price Stability Across Decades: Coefficient of Variation Analysis



KEY INSIGHTS

- Poetry, Sports & Recreation, Self-Help are most price-stable (CV <0.1) — strong reader expectations anchor prices over decades.
- Technical/specialised categories (Design, Art, Engineering) fluctuate dramatically — production costs and niche audience variability drive instability.
- Price ≠ Stability: Young Adult Fiction (mean $14.73, CV 0.07) is cheap and stable; Design (mean $18.41, CV 0.365) is similarly priced but volatile.
- Market maturity matters more than price level — stable categories have predictable, loyal buyer bases.

Author Gender Distribution Across Book Categories

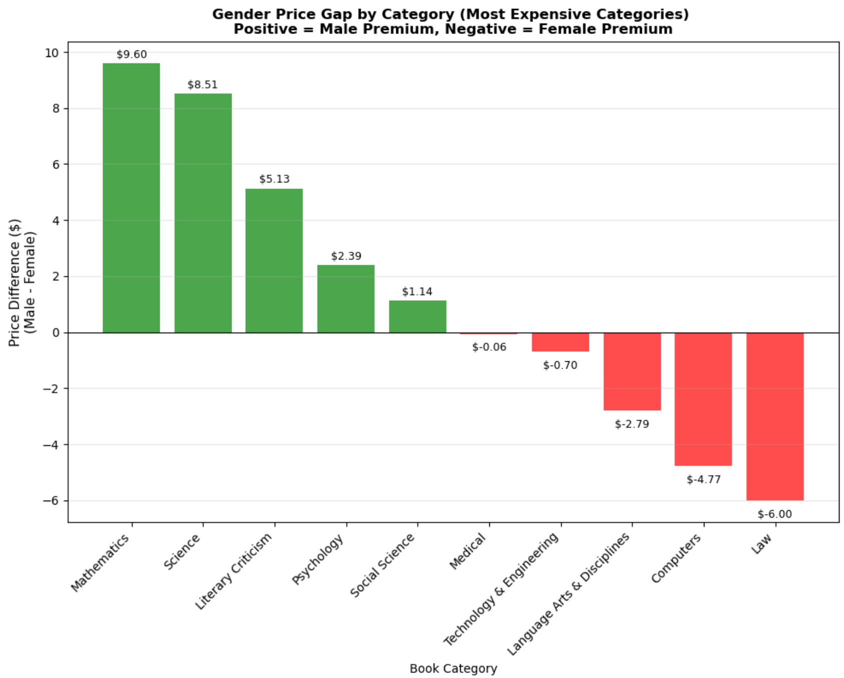


KEY INSIGHTS

- High-volume categories (Fiction, Juvenile Fiction, Family & Relationships) still show 3:1 male-to-female ratios despite appearing 'balanced'.
- Extreme polarisation in niche categories: 'Books and Reading' 0% female (777 books); 'Honesty' 100% female (127 books).
- Truly balanced categories (40–60% female) don't appear in top 10 by volume — parity correlates with niche, low-volume topics.
- Sample size critical: gender conclusions from small-count categories require cautious interpretation.

Gender Price Gap: Statistical Significance Analysis

Most Expensive Categories

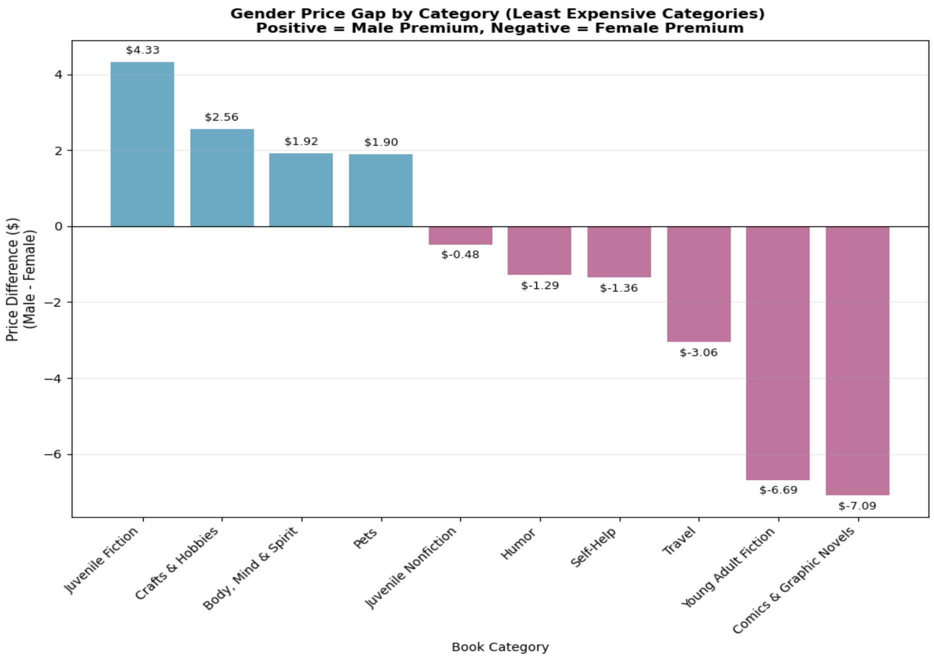


KEY INSIGHTS

- Mathematics: largest male premium (+$9.60, p=0.000) — highly significant, reflects extreme gender imbalance (694M vs 108F).
- Scarcity premium: female authors command higher prices in Computers (+$4.77), Law (+$6.00), Language Arts (+$2.79) — all statistically significant.
- Science gap ($8.51) appears large but is NOT significant (p=0.382) — only 46 female authors vs 1,995 male. Sample size is the story.
- Positive bars = male premium. Negative bars = female premium. P-values distinguish real market patterns from chance.

Gender Price Gap: Statistical Significance Analysis

Least Expensive Categories



KEY INSIGHTS

- In affordable categories, female authors command higher prices in the majority of cases (6 out of 10 categories), particularly in youth entertainment (Young Adult Fiction, Comics) and self-improvement niches. The male premium is mainly in traditional/children's categories.
- Sample size skews significance. Travel shows -$3.06 gap but insignificant (p=0.549) due to tiny female count (87 vs 648 males), while Crafts (+$2.56) also insignificant despite decent samples.
- Statistical significance ≠ practical significance: some large gaps are noise from tiny samples.

Key Findings Across All Three Research Questions

Q1 Market Evolution

- 1990s = inflection point: digital reviewing created a 10× signal shift in popularity data
- Fiction is the defining category of modern publishing — dominant from 1930s to present
- Classic literature can endure: Jane Eyre achieves genuine modern engagement across two decades

Q2 Price & Stability

- Technical/academic categories command 3× the price of leisure content — a persistent, structural premium
- Price stability is driven by market maturity, not price level itself
- Poetry and Self-Help are among the most stable categories — strong reader conventions anchor prices

Q3 Gender Dynamics

- Female authors command scarcity premiums in high-demand technical fields (Computers, Law)
- Most gender-balanced categories by percentage are niche and low-volume
- Sample size is the critical lens — statistical significance testing revealed several 'large' gaps are noise

Dataset, Methodology & Interpretive Constraints

Category Inconsistencies

Poor standardisation across decades — duplicates ('Bibles' vs 'Bible'), misclassifications (The Jungle Book under 'Animals'), and inconsistent tagging affect cross-decade trend reliability.

Author Gender Approximation

Genderize.io uses first-name inference — prone to cultural bias, exclusion of non-binary identities, and error on ambiguous names. Limited to top 30 authors only.

Popularity Metric Design

Composite score (review volume × rating) favours post-1990 books. Pre-digital books are systematically underrepresented regardless of historical significance.

Temporal Data Sparsity

Uneven data density across decades — sparse pre-1950s samples create noise in early decade comparisons. Results for older decades should be interpreted cautiously.

Price Sample Size Issues

Several categories have very small female author counts — apparently large gender price gaps are statistically insignificant. Always read alongside p-values.

Category Evolution Over Time

Static labels don't capture how genre definitions evolved (e.g., 'Self-Help' in 1940 vs 2020). Cross-decade category comparisons assume definitional consistency that may not hold.

Tools, Stack & Reproducibility

Python

Pandas, NumPy — data cleaning, transformation, and preprocessing pipeline across 700k+ records

DuckDB

Columnar analytical database — all aggregations and joins executed via SQL; no pandas post-load

Matplotlib / Seaborn

All visualisations — custom styling, statistical annotations, and multi-panel comparative figures

Genderize.io API

REST API integration — first-name gender inference with confidence scoring for top 30 authors

Kaggle Datasets

Amazon Books (212k) and Reviews (480k) — sourced, validated, and version-locked

Jupyter Notebooks

Full pipeline documented end-to-end; modular cells per research question for reproducibility

Dataset: https://www.kaggle.com/datasets/mohamedbakhet/amazon-books-reviews